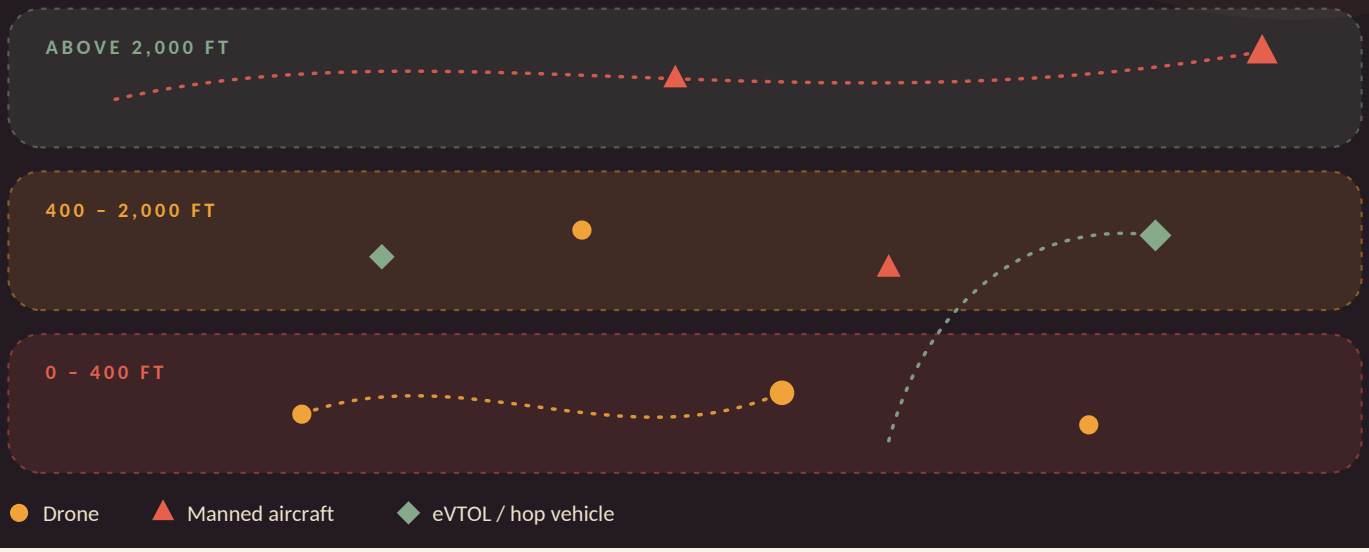
PROJECT PROPOSAL

DWBI Lab · Phase 1 · October 2026

AeroGrid

Low-Altitude Airspace Intelligence Warehouse



THE PROBLEM

Drones, small aircraft and new air taxis now share the sky close to the ground. The people who manage that space cannot easily see how crowded it is, where vehicles come too close, or how rules and weather change flying. AeroGrid brings live tracking, airspace rules, weather and flight records into one data warehouse so these questions can be answered with data.

WHAT WE AIM TO DO

01 One clear view
A single, trusted picture of low-altitude traffic.

02 Measure crowding
By altitude layer, map cell and time of day.

03 Find the risk
Detect and count close encounters between vehicles.

04 Plan ahead
See what rules and weather change, and where new drone corridors fit.

THE DATA

Five sources, one warehouse

OpenSky Network

Live and past aircraft positions: location, height, speed, direction.

JSON · CSV | every 5 min

FAA UDDS

Drone-flight maps, flight restrictions and airspace zone boundaries.

GeoJSON · Shapefile | weekly

TartanAviation (CMU)

Detailed flight paths recorded near airports over many months.

CSV · TXT | one-time load

Aviation Weather (NOAA)

Wind, visibility and cloud ceiling from airport stations.

JSON · CSV · METAR | hourly

Drone telemetry + generated paths

Pure drone flight logs and made-up short-hop paths that mimic eVTOL vehicles.

CSV · JSON | batch loads

HOW IT FLOWS

Bronze to Gold, on autopilot

BRONZE

Raw staging

Data lands exactly as received.

→

SILVER

Cleaned

Units fixed, duplicates removed, positions matched to zones.

→

GOLD

Star schema

Facts and dimensions, ready for analysis.

Automated: GitHub Actions runs scheduled Python jobs. Timestamp watermarks make sure nothing is loaded twice.

PROBLEMS WE ALREADY SOLVED IN THE DESIGN

Mixed units
Everything stored in feet.

Different IDs
One shared vehicle table for aircraft and drones.

Weather vs. map
Matched by nearest station and hour.

Zones
Positions placed inside restriction areas.

THE PLAN

What we measure and ask

5	5	5	5	10
sources	business processes	fact tables	dimensions	key questions

FIVE THINGS HAPPENING IN THE SKY

1	**Vehicle tracking**	Every position report from every vehicle.	`FactVehicleTracking`
2	**Flight journey**	A whole trip, from take-off to landing.	`FactFlightJourney`
3	**Restriction active**	When a no-fly rule starts, ends and where.	`FactRestrictionActivation`
4	**Close encounter**	Two vehicles closer than a safe distance.	`FactCloseEncounter`
5	**Airspace occupancy**	Regular snapshots of how full each cell is.	`FactAirspaceOccupancy`

Shared dimensions: Time · Vehicle · Spatial Grid · Weather · Airspace Zone

TEN QUESTIONS THE WAREHOUSE WILL ANSWER

1. When and where do altitude layers get crowded?
2. Which parts of the sky stay busy the longest?
3. How do traffic and flying height change by month and season?
4. When does traffic peak, and how does the drone vs. manned mix shift?
5. How many close encounters happen each day, and where?
6. Which vehicle types appear most in close encounters?
7. How much does traffic drop near an active restriction?
8. Do wind, low visibility or low clouds cut flights or raise risk?
9. What happens if drone numbers double or triple in a corridor?
10. Which quiet, safe zones could host new drone corridors?

THE OUTCOME

A star-schema warehouse fed by a live, automated pipeline, built on public data, that helps traffic managers, regulators, drone operators and city planners see, measure and plan the low-altitude sky.